

PARITY — Review III cheat sheet

BCSE497J · 16 September 2026 · Implementation · Technical accuracy · Results · Clarity

The 40-second version. Horror antagonists are static (learnable, stop being frightening) or random (unlearnable, feel unfair). PARITY builds an online behavioural model of the individual player and acts on it — intercepting where they are going, and degrading the information they depend on. The game is the application: the AI lives inside a playable first-person build and narrates its own reasoning on screen.

What this panel asked for last time, and what changed. "See the AI functioning" → it narrates every decision in English, in the game. "The game should BE the implementation" → the four prototypes are merged into one build. "About half the game" → the whole core loop is playable; the engine port is what remains. "Explain the AI in plain view" → an AI view draws its belief on the map and shows why it moves.

Numbers to know cold

RESULT	VALUE	THE POINT
Next-move accuracy, habitual player	83.6% vs 60.0% (order-1) vs 39.1% (none)	It learns routes
Same, random player (<i>the control</i>)	26.7% vs 26.9%	Identical — no leakage
How well trained, vs the ceiling	89.0% against a 91.8% oracle ceiling	97% of achievable signal; 100% vs a consistent player
Captures / 1000 moves	random 95 · hub-camp 301 · order-1 342 · VOMM 350	Any model beats none
Advantage vs facility size	1.22× at 10 rooms → 1.40× at 32	Prediction matters more at scale
World editing (no entity)	+63% cost per objective	No run ever soft-locked
Counter-play, 60% of doors wired	waiting +80% · disabling 3 sensors -21%	Killing sensors works, waiting does not
Predicting WHEN you stop to write	recall 17% vs 0% baseline	The open problem

10 rooms, 1200–1500 windows × 12 seeds, prequential. Training figures on 12 facilities never used to choose anything. 9 invariants, 23 commits.

Questions you will get

Where is the game?

Open `parity-3d.html`. First person, eight cells, three codebooks then the uplink, with the antagonist in the world hunting you. The systems view in `parity-game.html` is the same AI with its belief drawn on the map.

What is a VOMM?

Variable-order Markov model. It predicts your next move from your recent path, using as much history as it has reliable data for and backing off when it doesn't. Order-1 can't represent routes — a route means the same room has different successors depending on how you entered it.

How do you know it's trained well?

I measured the ceiling. An oracle that sees the player's own internal state reaches 91.8%; the rest is their coin-flips. The model reaches 89.0% from behaviour alone — 97% of what's achievable — and 100% against a perfectly consistent player. Grid search, intent tagging and goal inference all failed to improve it.

How do you know it isn't cheating or overfitting?

Two ways. Evaluation is prequential — every prediction precedes its outcome, so no leakage by construction. And the control: against a random player it scores the same as no model at all.

Can the player beat it?

Yes, but not by waiting — waiting is catastrophic (+439% at full coverage). The counter-play is spending a dangerous task to kill a sensor on the route you already use: -21% at 60% coverage. It backfires at full coverage, so it has a sweet spot.

Is it half the game?

The whole core loop is playable: move on a shared door clock, read codebooks, write pages, stage and send a byte, evade an AI that models you — in first person. What remains is the engine port, audio and content. Not mechanism.

Why Markov and not deep learning?

Cold start. It has to be interesting in the first two minutes against a player it has never seen, with no corpus for that individual. VOMM learns online from the first move, persists, and is inspectable — which is what makes the on-screen explanation possible.

Concede these fast, don't defend them. (1) No human playtest data — everything is against scripted players. (2) Predicting when you stop to write is weak (17% recall) and four attempts to fix it made it worse. (3) The first-person build is hand-written WebGL, not an engine; the Godot port hasn't started. (4) VOMM and order-1 are within noise on capture rate — the clear gap is at prediction, not pursuit.

Demo running order

#	WHAT	SAY
1	parity-3d.html	Walk two rooms. Let a window open and close. Let them read the panel top-right while you move.
2	parity-game.html	Same AI, systems view: belief as heat, solid vs dashed doors, and a sensor you can cut. Toggle AI view off, then on — that contrast is the demonstration.
3	python3 model/test_invariants.py	Nine invariants, a few seconds. Offer it if they press on rigour.